Yushan Dismitha Hettiarachchi

Undergraduate

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Profile

Computer Science undergraduate with hands-on experience in backend development and AI-powered application development using Java, Spring Boot, Python, and FastAPI. Skilled in REST API development, database integration, and LLM-based solutions, with project experience in AI-driven learning platforms and task management systems. Interested in building scalable backend systems and integrating AI technologies to solve real-world problems.

Education

BSc (Hons) Special Degree in Computer Science (GPA-3.60)
University of Ruhuna
15/08/2021 - up to date

Full-stack Master Development Program (Diploma)
Developer Stack
04/01/2025-05/01/2026

Advanced Level (1-A, 2-C)
D.S Senanayake College, Colombo
01/01/2018 – 15/11/2020

Projects

Thaksalawa-AI - Group Project (Team of 3 members)
AI powered LMS for students and teachers.
Technologies used - python (Fast API) , Langchain(RAG), Milvus, React (Frontend).
-Built a Vanilla RAG pipeline that ingests textbooks PDFs into Milvus Vector database (cloud) and powers a chatbot student questions grounded in lesson content. Built PDF ingestion workflow including text extraction, chunking and embedding generation.
-Conducted a comprehensive study on selecting best chunk size (rang 200 – 1200 tokens) for the embedded vector to provide a detailed answer. Experiment conducted on Grade 10 Science and History text books. With fixed chunk overlap 200. Finalized size of 1200 token considering number of chunks created and creation time for selected pdfs.
-Developed token based authentication system using JWT Bearer tokens
-Developed user management system with Python backend and MYSQL database. (admin,teacher,student creation,deletion and listing)

SmartTask – Individual Project
SmartTask is a task management platform with AI-powered features designed to improve productivity, organization, and workflow efficiency. The project combines a modern user experience with intelligent task planning and management capabilities to streamline daily work and collaboration.
Tech Stack – React.js, Spring Boot, FastAPI, Python, MYSQL, JWT Bearer tokens, Hugging face API, Meta Llama 3.1, Google SMTP(for reset functionality)

RIUSS – Group Project (Team of 4 members)
As the Frontend Developer for the Ruhuna International Undergraduate Science Symposium (RIUSS 2025), I designed and implemented a professional academic platform for the University of Ruhuna. I focused on creating a high-performance, responsive interface that serves as the central information hub for global researchers and students. The project required rigorous attention to detail—ensuring accessibility across all devices, managing complex scheduling layouts, and maintaining a visual identity aligned with university standards.

Projects

CRAG-Corrective Retrieval-Augmented Generation - Individual Project
Developed a Python-based question-answering application with a FastAPI backend and Streamlit frontend.
Implemented a LangGraph workflow that retrieves and evaluates document relevance, rewrites queries when necessary, performs web searches using Tavily, and generates context aware responses through an LLM.
Integrated ChromaDB and Hugging Face embeddings for PDF document storage and semantic retrieval.
Tech stack – Python, FastAPI, Streamlit, LangGraph, Tavily web search, OpenRouter API (

Infocure website – Group Project (Team of 5 members)
Developed a full-stack Spring-boot web application, with an admin dashboard for photo progress management and event handling and client side dashboard for event bookings.
Tech stack - Spring Boot, React, Tailwind css, SendGrid email services, Amazon S3 bucket, MongoDB, Javascript, Bitbucket version controlling
-Built Spring boot backend from scratch (Acted as the backend developer)
-Integrated S3 bucket for storing user payment slips.
-Implemented email service using SendGrid for password reset and final photos sends to the user.
-Built Session based authentication system.

Volunteering

AIESECER in University of Ruhuna
Manager - Outgoing Global Volunteer (OGV CXP)
01/02/2023 - 31/07/2025

Other Qualifications

Participated in Road to Insurgex 1.0 Hackathon
(Organized by CSSC University of Ruhuna)

Python Programming – stage 2
University of Moratuwa
August - 2026

Assured Diploma in English (DIE)
June - 2022

Skills

- **Programming Languages:** Java / Python / JavaScript / C
- **Databases**: MYSQL, MongoDB
- **Operating Systems:** Windows, Linux
- **Frontend Development:** React, NextJs
- **Backend Development:** Spring Boot, FastAPI
- **AI & Data Engineering:** RAG, Milvus, LLMs, Prompt Engineering, Vector Database
-**DevOps & Tools:** Git,GitHub,Docker,Postman,Render,Vercel